CervixScan · AI-Powered Cervical Cancer Screening · Project & Technical Overview

VGG16 Transfer Learning | TensorFlow Lite | Flask REST API

OVERVIEW

CervixScan is an open-source deep learning system that classifies cervical colposcopy images into Type 1, 2, or 3 transformation zones, helping clinicians and researchers make faster, more informed decisions. A finetuned-trained VGG16 CNN is served via a Flask REST API and accessed through a responsive web application.

KEY CAPABILITIES

→ 3-class cervical type classification
→ Real-time inference via REST API POST /classify
→ Drag-and-drop image upload with live preview
→ Per-class softmax confidence probability
→ Lightweight TFLite model

DATASET

Intel MobileODT Cervical Cancer Dataset

DATA PROCESSING

· Resize all images to 180 × 180 pixels (RGB)
· Normalize pixel values to range [0.0, 1.0]
· Detect & remove corrupt / truncated image files·
· Stratified train / val / test split (80% / 10% / 10%)
Rotation ±40° · Zoom 20% · Width shift 20%
· Shear 20% · Horizontal & Vertical flip enabled

MODEL ARCHITECTURE

VGG16 Base (ImageNet weights); Custom Softmax Head

Input Layer
180 × 180 × 3 (RGB)

VGG16 Conv Base
Output 5×5×512 · 14.7M total params · Last 5 layers unfrozen

Flatten
12,800 units

Dropout (p = 0.5)
Regularization — reduces co-adaptation

Dense (Softmax)
3 output neurons → Type 1 | Type 2 | Type 3 · 38,403 params

Training Configurations

Optimizer	**Adam (learning rate = 0.0001)**
Loss	**Sparse Categorical Cross-Entropy**
Batch Size	**32**
Max Epochs	**100 (early stopped at epoch 52)**
LR Schedule	**ReduceLROnPlateau (factor 0.2, patience 10)**
Early Stop	**Patience 20, restore best weights**
Checkpoint	**ModelCheckpoint (monitors val_loss)**

TRAINING & EVALUATION RESULTS

71.3%	72.4%	70.6%
Test Accuracy	Precision	Recall

DEPLOYMENT

GitHub Repository
Full source code including model training notebook, Flask API backend, and responsive frontend. Open for inspection and reproducibility.
github.com/Samuel-Ozechi/CervixScan

Web Application
Responsive SPA with drag-and-drop upload, live image preview, real-time classification, and per-class animated confidence bars.
cervixscan.com

IMPACT

Critical Screening Gap in Sub-Saharan Africa
Cervical cancer is the leading cause of cancer-related death among women in Sub-Saharan Africa. Providing automated diagnostics at the point of care specialists and makes essential screening accessible.

Academic Citation · Peer-Reviewed Publication
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"A regularized CNN approach for detecting cervical cancer"
https://pubs.aip.org/aip/acp/article-abstract/3220/1/030002/3315899/